Digital Image Processing (CS-323)

Lecture-4

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Outline

- Topics to Learn

1. Relationship between Pixels
   - Adjacency, connectivity, Regions, and Boundaries
2. Imaging Geometry

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Relationship between Pixels

- An image is denoted by *f(x, y)*, and for representing a pixel, we have *p* and *q*.
- A pixel *p* at coordinates (x, y) has four horizontal and vertical neighbors.

- The set of pixels termed as 4- neighbors of p is denoted by $\boldsymbol{N_4(p)}$ as illustrated below.

(x+1, y), (x-1, y), (x, y+1), and (x, y-1)

- The four diagonal neighbors of p is denoted as $\boldsymbol{N_D(p)}$ as

(x+1, y+1), (x+1, y-1), (x-1, y+1), and (x-1, y-1)

- Both $\boldsymbol{N_4(p)}$ and $\boldsymbol{N_D(p)}$ together referred as 8-neighbour of *p* denoted as $\boldsymbol{N_8(p)}$.

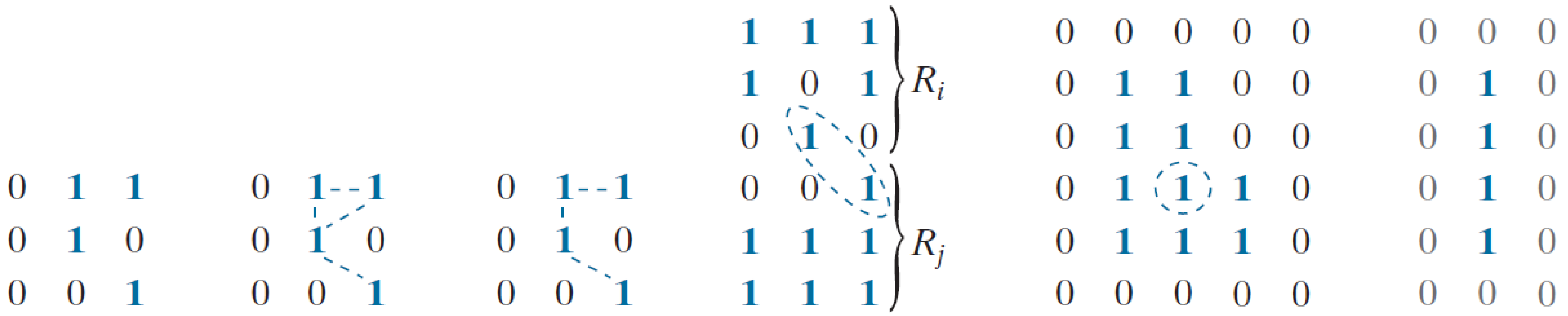
Adjacency, Connectivity, Regions and Boundaries

- Let V be the set of intensity used to define adjacency.
- In binary image $V = \{1\}$, we refers adjacency of pixels with value 1.
- For gray-scale image, V contain more elements, possible 256 values.
- There are three type of adjacency:

1. ***4-adjacency***: Two pixels p and q with values from V *are 4-adjacent* if q is in set $\boldsymbol{N_4(p)}$.
2. ***8-adjacency***: Two pixels p and q with values from V are *8-adjacent* if q is in set $\boldsymbol{N_8(p)}$.
3. ***m- adjacency (mixed adjacency)***: Two pixels p and q with values from V *are m-adjacent* if :
   I. If q is in $\boldsymbol{N_4(p)}$ or
   II. q is in $\boldsymbol{N_D(p)}$ and the set $\boldsymbol{N_4(p)} \cap \boldsymbol{N_4(q)}$ has no pixel whose value are from V.

Introduced to eliminate the ambiguities that arise when 8-adjacency is used.

Adjacency, Connectivity, Regions and Boundaries



a b c d e f

FIGURE 2.28 (a) An arrangement of pixels. (b) Pixels that are 8-adjacent (adjacency is shown by dashed lines). (c) m-adjacency. (d) Two regions (of 1's) that are 8-adjacent. (e) The circled point is on the boundary of the 1-valued pixels only if 8-adjacency between the region and background is used. (f) The inner boundary of the 1-valued region does not form a closed path, but its outer boundary does.

(Courtesy: Digital Image Processing, Gonzalez 3rd edition)

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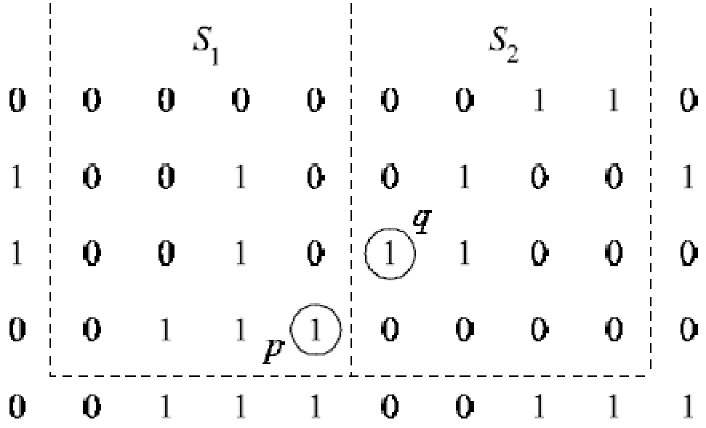
Adjacency, Connectivity, Regions and Boundaries: Numerical

- Consider the two image subsets, S1 and S2 in the following figure, assuming that V = {1}, determine whether these two subsets are:

  (a) 4-adjacent.

  (b) 8-adjacent.

  (c) m-adjacent.



(Courtesy: Digital Image Processing, Gonzalez 3rd edition)

Solution: a) It is not 4-adjacent, q is not in $N_4(p)$.

b) Yes, it is 8-adjacent, as q is in $N_8(p)$

c) Yes, it is m-adjacent as q is in $N_D(p)$ and $N_4(p) \cap N_D(p)$ is empty.

Adjacency, Connectivity, Regions and Boundaries: Numerical

- let V = {0,1} be the set of intensity values used to define adjacency.

Compute the lengths of the shortest 4-, 8-,and m-path between p and q in the following image. If a particular path does not exist between these two points, explain why.

Repeat for V = {1, 2}

	3	1	2	1 (q)
	2	2	0	2
	1	2	1	1
(p)	1	0	1	2

Solution:

For V = {0,1},

a) For 4-adjacency, there is no path, b) With 8-adjancency the shortest length is 4. c) With m-adjacency, the shorted path is 5.

Note: Both 8 and m adjacency path are unique.

For V = {1,2},

a) For 4-adjacency, shortest path length is 6, b) With 8-adjancency the shortest length is 4 (not unique). c) With m-adjacency, the shorted path is 6 (not unique).

Image Geometry

- Spatial information are performed directly on the pixels of the given image.
- It is classified on the basis of spatial operations-

1. Single-pixel operation, 2. Neighborhood operations, 3. Geometric spatial transforms.

- **Single-pixel operation:** These operations are performed to alter the values of individual pixels based on their intensity. Say, **S= T(z)**
- Where z is the intensity of the pixel of original image and s is mapped intensity of the corresponding pixel in the processed image.

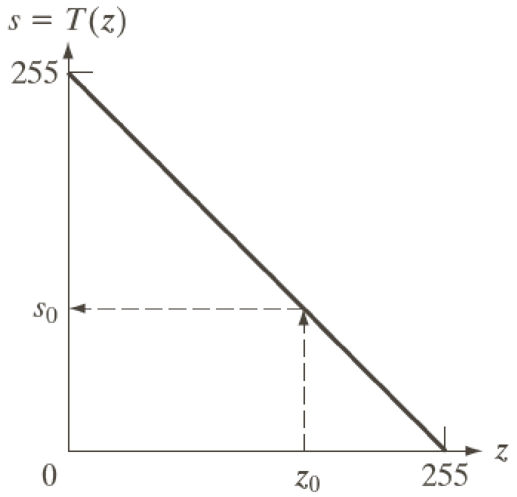


FIGURE 2.34 Intensity transformation function used to obtain the negative of an 8-bit image. The dashed arrows show transformation of an arbitrary input intensity value z_0 into its corresponding output value s_0.

(Courtesy: Digital Image Processing, Gonzalez 3rd edition)

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Image Geometry

- **Neighborhood operation:** Let S_{xy} denote the set of coordinate of a neighborhood centered at an arbitrary point (x, y) in an image , f.

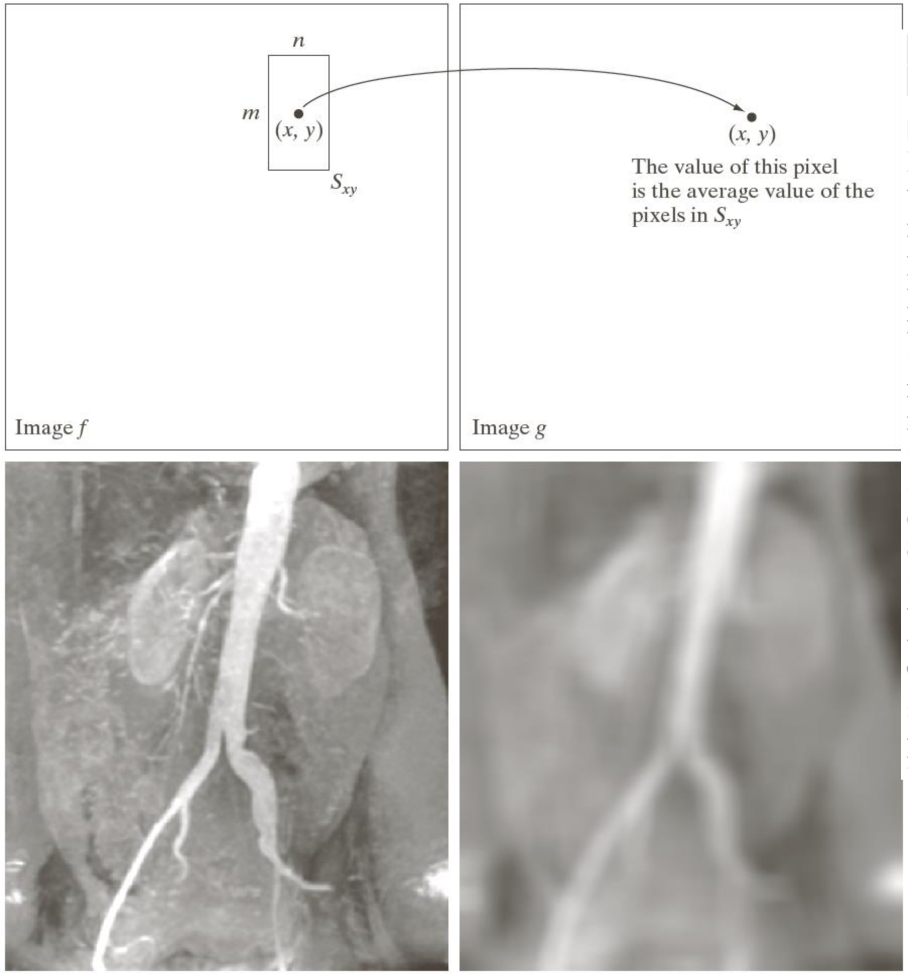
- Neighborhood processing generates a corresponding pixel at the same coordinate in an output (processed) image, g.

- Say, to compute average of the pixel in a rectangular neighborhood *mxn* centered on (x, y) is given as:

$$G(x, y) = \frac{1}{mn} \sum_{(r,c)\epsilon S_{xy}} f(r, c)$$

Image Geometry

- The net effect is to perform local blurring in the original image.
- This eliminate small details.



a b
c d

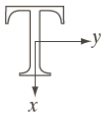
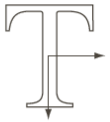
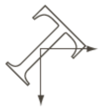
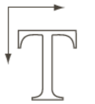
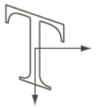
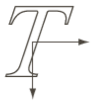
FIGURE 2.35 Local averaging using neighborhood processing. The procedure is illustrated in (a) and (b) for a rectangular neighborhood. (c) The aortic angiogram discussed in Section 1.3.2. (d) The result of using Eq. (2.6-21) with $m = n = 41$. The images are of size 790×686 pixels.

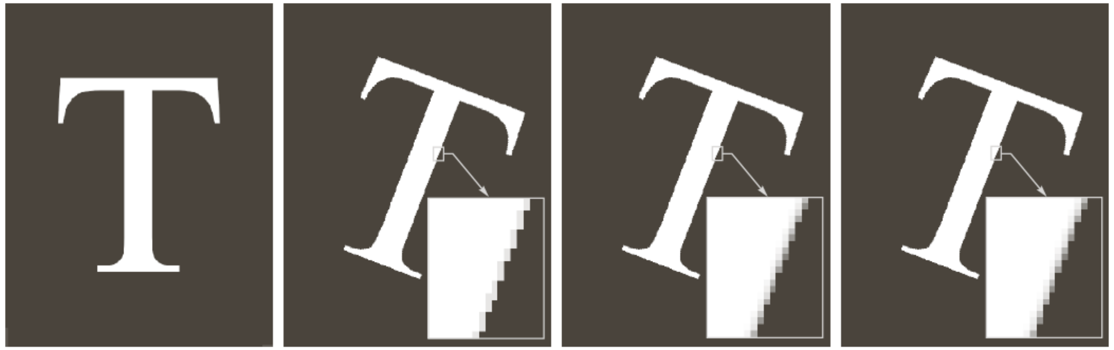
(Courtesy: Digital Image Processing, Gonzalez 3rd edition)

Image Geometry

- **Geometric image transform:** It modify the spatial relationship between pixels in an image (also called rubber-sheet transform).
- Example of spatial trans form are:
- Sub-sampling, identity, rotation, translation and so on.

TABLE 2.2
Affine transformations based on Eq. (2.6.–23).

Transformation Name	Affine Matrix, T	Coordinate Equations	Example
Identity	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x = v$ $y = w$	
Scaling	$\begin{bmatrix} c_x & 0 & 0 \\ 0 & c_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x = c_x v$ $y = c_y w$	
Rotation	$\begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x = v\cos\theta - w\sin\theta$ $y = v\cos\theta + w\sin\theta$	
Translation	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ t_x & t_y & 1 \end{bmatrix}$	$x = v + t_x$ $y = w + t_y$	
Shear (vertical)	$\begin{bmatrix} 1 & 0 & 0 \\ s_v & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x = v + s_v w$ $y = w$	
Shear (horizontal)	$\begin{bmatrix} 1 & s_h & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x = v$ $y = s_h v + w$	



a b c d

FIGURE 2.36 (a) A 300 dpi image of the letter T. (b) Image rotated 21° clockwise using nearest neighbor interpolation to assign intensity values to the spatially transformed pixels. (c) Image rotated 21° using bilinear interpolation. (d) Image rotated 21° using bicubic interpolation. The enlarged sections show edge detail for the three interpolation approaches.

(Courtesy: Digital Image Processing, Gonzalez 3rd edition)